MATH458 - Differential Geometry

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§1 Some Review

We will work in $\mathbb{R}^n$, usually with $n = 2, 3$. For vectors $v = (v_1, \ldots, v_n), w = (w_1, \ldots, w_n) \in \mathbb{R}^n$, we denote the dot product

$$v \cdot w = \sum_{i=1}^{n} v_i w_i.$$

More generally, an *inner product* on $\mathbb{R}^n$ is any function $b : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ that is symmetric, bilinear and positive definite. For instance, if $T : \mathbb{R}^n \to \mathbb{R}^n$ is linear and invertible $b_T(v, w) := T(v) \cdot T(w)$ a new inner product. In fact, it turns out every inner product on $\mathbb{R}^n$ is of this form; this implies that every inner product is just a coordinate-change away from the dot product.

We will say a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ is *orthogonal* if it is inner product preserving, i.e. $T(v) \cdot T(w) = v \cdot w$ for every $v, w \in \mathbb{R}^n$.

Exercise 1.1: Show that T is inner product preserving iff it is norm preserving ($\|Tv\| = \|v\|$) iff it is distance preserving ($\|T(v - w)\| = \|v - w\|$).

Exercise 1.2: Show that if T orthogonal, it is a bijection with determinant ± 1.

We say $T : \mathbb{R}^n \to \mathbb{R}^n$, linear, is *orientation preserving* if $\det(T) > 0$.

↪**Definition 1.1** (Rigid Motion): A function $M : \mathbb{R}^n \to \mathbb{R}^n$ is called a *rigid motion* if there exists an $a \in \mathbb{R}^n$ and $T : \mathbb{R}^n \to \mathbb{R}^n$ orthogonal and orientation preserving such that

$$M(v) = a + Tv, \quad \forall v \in \mathbb{R}^n.$$

We view the space $\mathbb{E}^n$ as $\mathbb{R}^n$ equipped with the Euclidean distance, which we'll denote $d_{\mathbb{E}}$ or d if no confusion arises, *up to rigid motions*. In practice, this means working in $\mathbb{E}^n$ has no distinguished origin point or coordinate axes. However, also in practice, we will make the identification $\mathbb{E}^n \simeq \mathbb{R}^n$ by picking an origin and axes, as we will see.

However, working in $\mathbb{E}^n$, abstractly, still preserves orientation and distance, since these are both preserved under rigid motions.

For $r > 0$ and $\rho \in \mathbb{E}^n$, we write $\mathbb{D}_r(\rho)$ for the open unit disk, and $\mathbb{D}^n := \mathbb{D}_1(0) \subset \mathbb{R}^n$.

↪**Theorem 1.1** (Heine-Borel): $C \subset \mathbb{E}^n$ compact iff closed and bounded.

Exercise 1.3: Let $r' > r > 0$ and $\rho \in \mathbb{E}^n$. Let $f : \mathbb{D}_{r'}(\rho) \to \mathbb{E}^n$ be continuous. Show that $f|_{\mathbb{D}_r(\rho)}$ uniformly continuous.

We'll denote the derivative of a function $f : \mathcal{U} \subset \mathbb{R}^n \to \mathbb{R}^m$ at a point a by $D_a f : \mathbb{R}^n \to \mathbb{R}^m$, which is represented by the Jacobian $m \times n$ matrix $J(f)_a = \left(\frac{\partial f}{\partial x_1}|_a, \ldots, \frac{\partial f}{\partial x_n}|_a \right)$.

↪**Definition 1.2**: We will say $f : \mathcal{U} \to \mathbb{R}^m$ is C^k on $\mathcal{U}$ if all the kth order partial derivatives of all of the component functions of f are continuous. We say f in C^∞ if it is in C^k for every $k \geq 1$. We write C^0 for the space of continuous functions.

Remark 1.1: $C^{k+1} \Rightarrow C^k$

§2 Curves

↪**Definition 2.1** (Parametrized curve/path): A *parametrized curve/path* in $\mathbb{E}^n$ is a continuous function

$$\gamma : I \to \mathbb{E}^n,$$

where $I \subset \mathbb{R}$ an interval. We say γ *compact* if I is compact.

↪**Definition 2.2** ((Regular) C^k parametrized curve): Fix coordinates in $\mathbb{E}^n$. Then, a (regular) C^k parametrized curve is a parametrized curve in which $\gamma \in C^k(I)$ (and for which $\frac{\mathrm{d}\gamma}{\mathrm{d}t}(t) \neq 0 \forall t \in I$).

Exercise 2.1: Regularity and differentiability is preserved under rigid motion, i.e. if γ a (regular) C^k parametrized curve and M a rigid motion on $\mathbb{R}^n$, then $\tilde{\gamma} := M \circ \gamma$ also (regular) C^k.

Solution. Indeed, if $M(x) = a + Tx$ then $M'(x) = T$ so that by chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}(M \circ \gamma)(t) = T \cdot \dot{\gamma}(t)$$

(this easily shows differentiability) and since T orthogonal, we find $\|T\dot{\gamma}\| = \|\dot{\gamma}\| > 0$. ■

↪**Definition 2.3**: Given a curve γ, we define

- the *velocity*, $v = \frac{\mathrm{d}\gamma}{\mathrm{d}t} : I \to \mathbb{R}^n$
- the *acceleration*, $\alpha = \frac{\mathrm{d}^2\gamma}{\mathrm{d}t^2} : I \to \mathbb{R}^n$
- the *speed*, $\sigma = \|v\| = \left\|\frac{\mathrm{d}\gamma}{\mathrm{d}t}\right\| : I \to \mathbb{R}$,

whenever each of these quantities all exist.

Exercise 2.2: Speed is preserved by rigid motions.

Solution. We basically showed this in the previous exercise. ■

↪**Definition 2.4**: Let γ be a C^1 curve. The *arclength* of γ is defined by

$$\ell(\gamma) := \int_I \sigma(t)\,\mathrm{d}t.$$

Exercise 2.3: Let $p, q \in \mathbb{E}^2$ with $d_{\mathbb{E}}(p,q) = 3$. Suppose $\gamma : [a,b] \to \mathbb{E}^2$ is a C^1-path with $\gamma(a) = p, \gamma(b) = q$. Prove that $\ell(\gamma) \geq 3$, with equality holding iff $\gamma(I)$ is a line segment, with no change of direction.

(Hint: pick coordinates so that $p = 0$ and the x-axis passes through q to simplify computations.)

↪**Definition 2.5** (Curve): A set $\mathcal{C} \subset \mathbb{E}^n$ is a *curve* if it is connected, and for every $p \in \mathcal{C}$, there exists a compact neighborhood N_p of p and a one-to-one, compact, parametrized curve $\gamma : I \to \mathbb{E}^n$ such that $\gamma(I) = \mathcal{C} \cap N_p$.

A curve is called C^k if there exists γ as in the definition which are now required to be C^k.

I.e., a general curve is everywhere locally a compact parametrized curve.

Remark 2.1: One can relax the one-to-one/compact conditions to obtain either a global compact parametrization (which may not be one-to-one) or a parametrized curve with $I = \mathbb{R}$ with $\gamma(I) = \mathcal{C}$ and γ is periodic.

§2.1 Classification Theorem for Curves

↪**Theorem 2.1** (Classification Theorem for Curves): Let $\mathcal{C} \subset \mathbb{E}^n$ a connected subset. Then, $\mathcal{C}$ is a (regular) $[C^k]$ curve iff it is the image of a (regular) $[C^k]$ path $\gamma : I \to \mathbb{E}^n$ satisfying either

1. γ is one-to-one with $[C^k]$ continuous inverse
2. $I = \mathbb{R}$ and γ is periodic, and the restriction of γ to any interval I' shorter than the period is one-to-one.

If γ satisfies 1. or 2., we'll call it a *global parametrization* of $\mathcal{C}$.

Remark 2.2: This means we just need *one* path to describe a curve; but it may, in 2., loop back onto itself.

§2.2 Reparametrizations of Curves

↪**Definition 2.6** (Reparametrization): Let $I, \tilde{I} \subset \mathbb{R}$ be intervals and $t : \tilde{I} \to I$ a continuous bijection (we'll call it a *change of parameters*). Then, the *reparametrization* of $\gamma : I \to \mathbb{E}^n$ using t is the composition $\tilde{\gamma} := \gamma \circ t : \tilde{I} \to \mathbb{E}^n$.

Suppose γ a regular C^k path and $t : \tilde{I} \to I$ a C^k bijection with a C^k inverse. Then $\tilde{\gamma}$ is a C^k-reparametrization of γ.

We say t is *orientation-preserving (-reversing)* if it is monotone increasing (decreasing).

Remark 2.3: γ also a reparametrization of $\tilde{\gamma}$ using the inverse $s := t^{-1}$.

↪**Theorem 2.2**: Suppose $\gamma : I \to \mathbb{R}^n$ is C^1 and $\tilde{\gamma} : \tilde{I} \to \mathbb{R}^n$ a C^1 reparametrization of γ. Then $\ell(\gamma) = \ell(\tilde{\gamma})$, that is, arclength is invariant under change of parameters.

↪**Theorem 2.3** (Arc-Length Parametrization): Let $\gamma : I \to \mathbb{E}^n$ be a regular C^k path. Then, there exists an orientation-preserving C^k reparametrization of γ, $\tilde{\gamma} : \tilde{I} \to \mathbb{E}^n$, with unit speed, i.e. $\|\dot{\tilde{\gamma}}\| \equiv 1$.

Proof. Pick $t_0 \in I$ and definte

$$s : I \to \mathbb{R}, \quad s(t) := \int_{t_0}^{t} \|\dot{\gamma}(r)\| \, \mathrm{d}r.$$

This integral exists and is bounded, and moreover,

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \|\dot{\gamma}(t)\| > 0,$$

since γ regular. In particular, we see that s is invertible on its image $\tilde{I} := s(I)$, and increasing. Then, $s : I \to \tilde{I}$ an orientation-preserving, C^1 bijection with $s' > 0$. By the inverse function theorem, $t := s^{-1} : \tilde{I} \to I$ exists and has the same desired properties. Moreover,

$$t'(s) = \frac{1}{s'(t(s))} = \frac{1}{\|\dot{\gamma}(t(s))\|}.$$

Letting $\tilde{\gamma} := \gamma \circ t$, then we see that

$$\|\dot{\tilde{\gamma}}(s)\| = \|\dot{\gamma} \circ t(s) \cdot t'(s)\| = \frac{1}{\|\dot{\gamma}(t(s))\|}\|\dot{\gamma}(t(s))\| \equiv 1.$$

■

Exercise 2.4: Any two arc-length parametrizations differ by some shifting in the domain, i.e. if $\gamma_i : I_i \to \mathbb{R}$ are two arc-length reparametrizations of a regular path $\gamma : I \to \mathbb{R}^n$ using a change of parameters $t_i : I_i \to I$ for $i = 1, 2$, then $h := t_2^{-1} \circ t_2 : I_2 \to I_1$ is a restriction of a rigid motion of $\mathbb{R}$; specifically $h' \equiv 1$.

With this, we can try to define the length of a general curve C. Suppose $C \subset \mathbb{E}^n$ a compact curve with boundary $\{p, q\}$ (so satisfies the first point of the classification theorem).

1. If C a line segment, then we just define

$$\mathcal{L}_1(C) := d_{\mathbb{E}}(p, q).$$

2. If C regular, then we define

$$\mathcal{L}_2(C) := \ell(\gamma),$$

   where γ is any parametrization of C.

Exercise 2.5: This definition of $\mathcal{L}_2$ is well-defined, i.e. independent of choice of parametrization.

↪**Definition 2.7** (Rectifiable): Let $\mathcal{C}$ be a compact curve with boundary $\{p, q\}$. An *inscribed polygon* in $\mathcal{C}$ is a finite increasing sequence of points $\mathcal{P} = \{p_i\}_{i=0}^{N}$ of points in $\mathcal{C}$ with endpoints $p_0 = p, p_N = q$. We write

$$L(\mathcal{P}) := \sum_{i=0}^{N-1} d_{\mathbb{E}}(p_i, p_{i+1})$$

for the length of $\mathcal{P}$, and

$$|\mathcal{P}| := \max_{i=0}^{N-1} d_{\mathbb{E}}(p_i, p_{i+1})$$

for the size of $\mathcal{P}$.

A curve $\mathcal{C}$ is said to be *rectifiable* if there exists a real number $\mathcal{L}_3(\mathcal{C}) \geq 0$ such that for all sequence $\{\mathcal{P}_m\}$ of inscribed polygons in $\mathcal{C}$ with $|\mathcal{P}_m| \underset{m\to\infty}{\to} 0$, we have

$$\lim_{m\to\infty} L(\mathcal{P}_m) = \mathcal{L}_3(\mathcal{C}).$$

↪**Proposition 2.1**: A unit-speed reparametrization is essentially unique, up to a shift in the domain I.

Exercise 2.6: Compute the arc-length parametrization of $\gamma(t) := (t, t^2)$.

↪**Lemma 2.1**: Let $\tilde{\gamma} : \tilde{I} \to \mathbb{R}^n$ be a regular C^2 path with constant speed. Then, $\ddot{\tilde{\gamma}}$ will always be orthogonal to $\dot{\tilde{\gamma}}$.

Proof. Suppose $\|\dot{\tilde{\gamma}}\| \equiv c$. We apply the product rule for dot products, to obtain

$$\begin{aligned} 0 = \frac{\mathrm{d}}{\mathrm{d}t}(c^2) &= \frac{\mathrm{d}}{\mathrm{d}t}\|\dot{\tilde{\gamma}}\|^2 \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\dot{\tilde{\gamma}} \cdot \dot{\tilde{\gamma}} \\ &= 2\ddot{\tilde{\gamma}} \cdot \dot{\tilde{\gamma}}, \end{aligned}$$

which gives the proof. ■

§2.3 Curvature

Let γ be a regular C^2-path $\gamma : I \to \mathbb{R}^n$, there exists an orientation-preserving change of parameters $t : \tilde{I} \to I$ such that $\tilde{\gamma} := \gamma \circ t : \tilde{I} \to \mathbb{R}$ has unit speed. Let $s := t^{-1} : I \to \tilde{I}$.

↪**Definition 2.8** (Curvature of a parametrized curve): Define the curvature of γ as above at some time $t \in I$ to be

$$\kappa_\gamma : I \to \mathbb{R}_+, \quad \kappa_\gamma(t) := \|(\ddot{\tilde{\gamma}} \circ s)(t)\|.$$

Exercise 2.7: Show that this definition is well-defined, i.e. independent of choice of unit-speed parametrization.

↪**Definition 2.9** (Curvature of a curve): Given a regular C^2 curve $\mathcal{C} \subset \mathbb{R}^n$, there exists (by the classification theorem) a global, regular, C^2 parametrization of $\mathcal{C}$, $\gamma : I \to \mathbb{R}^n$. For a point $p \in \mathcal{C}$, then, there exists some $t \in I$ such that $\gamma(t) = p$. Define, then, the curvature of $\mathcal{C}$ at p, then, to be the curvature of γ at time t.

Exercise 2.8: Show that this definition is well-defined, i.e., independent of choice of regular global parametrization. One will need to appeal to the inverse function theorem, to show that any two such parametrizations differ by an orientation-preserving change of parameters.

Exercise 2.9: Show that curvature of preserved by rigid motions of $\mathbb{R}^n$, i.e. given M a rigid motion of $\mathbb{R}^n$ and a regular C^2 curve γ, then

$$\kappa_{M \circ \gamma} = \kappa_\gamma.$$

Remark 2.4: In particular, this exercise gives that curvature is an *inherit property* of curves in $\mathbb{E}^n$, not just in $\mathbb{R}^n$.

Remark 2.5: The definition of κ_γ is a little bothersome in the sense that it requires computing an arc-length parametrization. The follow result shows how we can compute it regardless.

↪**Proposition 2.2**:

$$\kappa_\gamma = \frac{1}{\|\dot{\gamma}\|^2}\left\|\ddot{\gamma} - \frac{\ddot{\gamma} \cdot \dot{\gamma}}{\dot{\gamma} \cdot \dot{\gamma}}\dot{\gamma}\right\| = \frac{\|\ddot{\gamma}^\perp\|}{\|\dot{\gamma}\|^2},$$

where we use the "$\perp$" notation to indicate the orthogonal complement of $\ddot{\gamma}$ with respect to $\dot{\gamma}$.

Proof. I'll add it later. It's just repeated application of the chain rule and product rule.

■

Exercise 2.10: Compute the curvature of parabola $\mathcal{C} := \{(x, y) \mid y = x^2\} \subset \mathbb{R}^2$ at any point.

↪**Theorem 2.4**: The quantity $\frac{\|\ddot{\gamma}^\perp\|}{\|\dot{\gamma}\|^2}$ is preserved under reparametrization.

Remark 2.6: This is really more of a corollary of the previous proposition. Moreover, this implies that our definition of curvature is "correct" as a property of curves in $\mathbb{E}^n$ rather than just $\mathbb{R}^n$.

↪**Definition 2.10**: Let $\gamma : I \to \mathbb{R}^n$ a regular path. We define

- $T(t) := \frac{\dot{\gamma}(t)}{\|\dot{\gamma}(t)\|}$, then *unit tangent* at time t

If $\gamma \in C^2$,

- $N(t) := \frac{\ddot{\gamma}(t)^\perp}{\|\ddot{\gamma}(t)^\perp\|}$, the *unit normal* at time t
- the *osculating plane* at time t is the plane in $\mathbb{R}^n$ that contains the point $\gamma(t)$ and is spanned by $\{\dot{\gamma}(t), \ddot{\gamma}(t)\}$ (supposing $\kappa_\gamma \neq 0$)
- the *osculating circle* at time t as the circle laying in the osculating plane of radius $\frac{1}{\kappa(t)}$ and centered at $\gamma(t) + \frac{N(t)}{\kappa(t)}$
- the *evolute* of γ is the map

$$t \in I \mapsto \gamma(t) + \frac{N(t)}{\kappa(t)} = \text{center of oscualting circle at } t$$

Remark 2.7: $\ddot{\gamma}^\perp \neq 0 \Leftrightarrow \kappa_\gamma \neq 0 \Leftrightarrow \{\dot{\gamma}, \ddot{\gamma}\}$ a linearly independent set.

Exercise 2.11: A circle of radius r, i.e. the curve defined implicitly by $\{x^2 + y^2 = r^2\}$, has curvature $\frac{1}{r}$.

This exercise shows that the osculating circle at a point on a curve has the same curvature as the curve at that point.

Exercise 2.12: Suppose $n = 2$ and a curve is given *explicitly* by $y = f(x)$ where $f : \mathbb{R} \to \mathbb{R}$ sufficiently differentiable. Compute the curvature in terms of f and its derivatives. Do the same if the curve is given *implicitly* as the set of $(x, y) \in \mathbb{R}^2$ such that $g(x, y) = 0$ where $g : \mathbb{R}^2 \to \mathbb{R}$ sufficiently differentiable.

Fix now $n = 2$. Let $\gamma : I \to \mathbb{R}^2$ be a regular C^2 curve and fix $t \in I$. Let us assume (by changing coordinates if necessary) that $\gamma(t) = 0$ and the x-axis is parallel to $T(t)$, i.e. $T(t) = (1, 0)$. Then, we see that we may write

$$\frac{\ddot{\gamma}(t)^\perp}{\|\dot{\gamma}(t)\|^2} = \text{constant} \times (0, 1).$$

Specifically, the "constant" here is what we call the *signed curvature* of γ at time t, and is computed as :

↪**Definition 2.11** (Signed curvature): Let γ as in the above, then the *signed curvature* is the quantity

$$\kappa_\gamma^\pm(t) = \frac{1}{\|\dot{\gamma}(t)\|^2} \frac{\ddot{\gamma}(t) \cdot \dot{\gamma}(t)^*}{\|\dot{\gamma}(t)\|},$$

where we use the notation v^* as a rotation of $v = (v_1, v_2)$ by an angle of $\frac{\pi}{2}$, counter-clockwise, i.e. $v^* = (-v_2, v_1)$.

Exercise 2.13: $\kappa_\gamma^\pm(-t) = -\kappa_\gamma^\pm(t)$.

Exercise 2.14: Suppose $\gamma(t) = (x(t), y(t))$, then show

$$\kappa_\gamma = \frac{\dot{x}\ddot{y} - \ddot{x}\dot{y}}{\left(\dot{x}^2 + \dot{y}^2\right)^{3/2}}.$$

↪**Definition 2.12** (Angle function): Let $\gamma : I \to \mathbb{R}^2$ be a regular C^2 curve parametrized by arc length with basepoint $s_0 \in I$. We assume wlog $s_0 = 0$ (by translating if necessary) and that $\dot{\gamma}(0) = (1, 0)$ (by changing coordinates). We define the *angle function* of γ by

$$\theta : I \to \mathbb{R}, \quad \theta(0) = 0, \theta(s) := \int_0^s \kappa_\gamma^\pm(u)\, \mathrm{d}u.$$

In particular, $\frac{\mathrm{d}\theta}{\mathrm{d}s} = \kappa_\gamma^\pm(s)$.

Remark 2.8: We can view $s \mapsto \dot{\gamma}(s)$ as a new C^1-parametrized curve, in which case its arc length is given by

$$\int_0^s \|\ddot{\gamma}(u)\|\, \mathrm{d}u = \int_0^s \kappa_\gamma(u)\, \mathrm{d}u.$$

So, in a sense, the θ angle function is the "signed arc-length" of $\dot{\gamma}$, i.e. it accounts for backtracking.

Moreover, since we have an arc length parametrization, we know $\dot{\gamma}$ a unit vector, hence we can view the map $t \mapsto \dot{\gamma}(t)$ as a map from I to the unit circle in $\mathbb{R}^2$. Hence, θ is meant to capture the angle of this unit vector for any t, i.e. $\dot{\gamma} = (\cos, \sin) \circ \theta$.

↪**Theorem 2.5** (Fundamental Theorem of Plane Paths): Let $s_0 \in I$ be a given base point and let $\kappa : I \to \mathbb{R}$ be a C^{k-2} function ($2 \leq k \leq \infty$). Then, for each $p \in \mathbb{R}^2$ and $\theta_0 \in \mathbb{R}$, there is a unique regular C^k path $\gamma : I \to \mathbb{R}^2$, parametrized by arc-length, such that

1. $\kappa_\gamma^\pm = \kappa$,
2. $\dot{\gamma}(s_0) = (\cos(\theta_0), \sin(\theta_0))$,
3. $\gamma(s_0) = p$.

Remark 2.9: The choice of p, θ_0 just correspond to a translation, rotation (resp.) of $\mathbb{R}^2$ of our curve, i.e. this means our curve is uniquely determined up to rigid motion.

Remark 2.10: This essentially says that, given the curvature of a curve in the plane, we can reconstruct the curve.

Proof. We seek to find $\gamma : I \to \mathbb{R}^2$ and $\theta : I \to \mathbb{R}$ such that

$$\frac{\mathrm{d}\gamma}{\mathrm{d}s} = (\cos\theta, \sin\theta), \gamma(s_0) = p$$

and

$$\frac{\mathrm{d}\theta}{\mathrm{d}s} = \kappa, \theta(s_0) = \theta_0.$$

By the fundamental theorem of calculus, we know

$$\theta(s) = \int_{s_0}^{s} \kappa(u)\,\mathrm{d}u + \theta_0$$

is the unique solution for $\theta(s)$ with the given properties, which in turn implies

$$\gamma(s) = \left(\int_{s_0}^{s} \cos(\theta(u))\,\mathrm{d}u, \int_{s_0}^{s} \sin(\theta(u))\,\mathrm{d}u\right) + p,$$

which is again unique by FTC. ■

Remark 2.11: This theorem essentially says that a curve is uniquely determined by its signed curvature. However, the same is not true if we just take the curvature. For instance, the curves given explicitly by $y = x^3$, $y = |x|^3$ have the same curvature everywhere but clearly do not described the same curves.

A more abstract manner of characterizing the angle function for a more general curve is as follows. If $\gamma : I \to \mathbb{R}^2$ a regular, C^2 curve, then the angle function $\theta : I \to I'$ where I' some other interval of $\mathbb{R}$, is such that

$$T = \rho \circ \theta,$$

where $\rho : I' \to \mathbb{R}^2$ is the standard parametrization of the circle given by $\rho(\theta) := (\cos(\theta), \sin(\theta))$ and T the unit tangent vector viewed as a map $I \to \mathbb{R}^2$.

Exercise 2.15: Characterize regular C^2 curves in $\mathbb{E}^2$ with constant curvature.

Solution. Let γ be an arc-length parametrization of a regular C^2 curve $\mathcal{C} \subset \mathbb{E}^2$, so that $\kappa = \|\ddot{\gamma}\| = \ddot{\gamma} \cdot \dot{\gamma}^*$. If $\kappa \equiv 0$, then $\ddot{\gamma} \equiv 0$ so $\dot{\gamma}$ a constant vector and thus γ of the form $ct + b$ for some vectors c, b, i.e. γ just a line in $\mathbb{E}^2$.

If κ nonzero, then we can write

$$\ddot{\gamma} = \kappa\dot{\gamma}^* \Rightarrow \dot{\gamma} = \kappa\gamma^* \Rightarrow \begin{cases} \dot{x} = -\kappa y \\ \dot{y} = \kappa x \end{cases}.$$

Solving this system of equations gives that γ traces a circle. Thus, $\mathcal{C}$ must either be a line or a circle. ■

Exercise 2.16: Show that the signed curvature of γ is preserved under rigid motion, hence is well-defined as a property of a curve in $\mathbb{E}^n$. (Note that the signed curvature is the derivative of the θ function, hence it suffices to prove this property for θ)

§2.4 3-Dimensional Space Paths

We wish to derive an analogous "fundamental" result for curves in $\mathbb{R}^3$. However, we have no notion of "signed curvature" in this case. Moreover, as we'll see, we actually need a second "intrinsic" (called *torsion*) of the curve to uniquely identify it.

Fix $\gamma : I \to \mathbb{R}^3$ regular and C^2 and with strictly positive curvature (turns out, there's not much we can say when the curvature is 0).

Define as before

$$T := \frac{\dot{\gamma}}{\|\dot{\gamma}\|}, \quad N := \frac{\ddot{\gamma}^{\perp}}{\|\ddot{\gamma}^{\perp}\|}$$

the unit tangent and normal vectors. Remark that $T \cdot N = 0$. Since we are in $\mathbb{R}^3$, there exists a unique third vector, which we denote B and call it the *binormal* such that $\{T, N, B\}$ is an orthonormal, positively oriented basis (in the sense that the matrix consisting of columns T, N, B in that order is orthogonal with determinant 1) of $\mathbb{R}^3$, i.e.

$$B := T \times N.$$

The basis $\{T, N, B\} \subset \mathbb{R}^3$ is called the *Frenet frame* associated to γ.

We'll be interested in the dynamics of this frame, i.e. how T, N, B resp. change in time. We need to additionally assume $\gamma \in C^3$ for this, so that we may take derivatives of N. We'll also assume γ is parametrized by arc-length for convenience. We find that with these assumptions,

$$T = \dot{\gamma}$$
$$\Rightarrow \dot{T} = \ddot{\gamma} = \|\ddot{\gamma}\| N = \kappa N.$$

In addition,

$$\|B\| = 1 \Rightarrow \dot{B} \cdot B = 0$$

and

$$B = T \times N \Rightarrow \dot{B} = \dot{T} \times N + T \times \dot{N} = \kappa \underbrace{N \times N}_{=0} + T \times \dot{N} \Rightarrow \dot{B} \cdot T = 0,$$

hence $\dot{B}$ is simultaneously orthogonal to B and T, hence

$$\dot{B} = \text{const}(-N).$$

We call this constant the *torsion* τ of γ at time s, which is given by

$$\tau := -\dot{B} \cdot N.$$

Finally, to compute $\dot{N}$, we have that

$$\|N\| = 1 \Rightarrow \dot{N} \cdot N = 0$$
$$T \cdot N = 0 \Rightarrow 0 = \dot{T} \cdot N + T \cdot \dot{N} = \kappa \underbrace{\|N\|^2}_{=1} + T \cdot \dot{N} \Rightarrow T \cdot \dot{N} = -\kappa$$
$$B \cdot N = 0 \Rightarrow 0 = \dot{B} \cdot N + B\dot{N} = -\tau + B \cdot \dot{N} \Rightarrow B \cdot \dot{N} = \tau.$$

This implies

$$\dot{N} = -\kappa T + \tau B.$$

In summary, we can succinctly write

$$\begin{pmatrix} \dot{T} \\ \dot{N} \\ \dot{B} \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix} \qquad \text{(The Frenet equations).}$$

↪**Theorem 2.6** (Fundamental Theorem of Space Paths): Let $I \subset \mathbb{R}$ be an interval with basepoint $s_0 \in I$. Suppose $\tau : I \to \mathbb{R}$ is a C^{k-3} function and $\kappa : I \to \mathbb{R}_{>0}$ is a C^{k-2} function (where $3 \leq k \leq \infty$). Then, for any initial point $p_0 \in \mathbb{R}^3$, initial velocity $v_0 \in \mathbb{R}^3$, and initial normal direction $n_0 \in \mathbb{R}^3$ such that $\|v_0\| = \|n_0\| = 1$ and $v_0 \cdot n_0 = 1$, there is a *unique* regular C^k path $\gamma : I \to \mathbb{R}^3$ parametrized by arc length and satisfying:

1. $\kappa_\gamma = \kappa$,
2. $\tau_\gamma = \tau$,
3. $\gamma(s_0) = p_0$,
4. $\dot{\gamma}(s_0) = v_0$,
5. $\frac{\ddot{\gamma}(s_0)}{\|\ddot{\gamma}(s_0)\|} = n_0$.

Remark 2.12: The last three requirements say that this curve is uniquely defined up to rigid motion, hence unique in $\mathbb{E}^3$; translations will simply change the initial point p_0, and rotations will change the angles of v_0, n_0.

Proof. Remark that the Frenet equations are a system of (9) first order ODEs with given initial condition. The Picard-Lindelhoff theorem from ODEs says that there exist unique function $T, N, B : I \to \mathbb{R}^3$ satisfying the equations with $T(s_0) = v_0, N(s_0) = n_0, B(s_0) = v_0 \times n_0$. We need to show that these are the Frenet frame of some curve.

First, we show they are a positively oriented orthogonal basis. Indeed, remark that, using the Frenet equations,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}s}(T \cdot N) &= \kappa(N \cdot N) - \kappa(T \cdot T) + \tau(T \cdot B) \\
\frac{\mathrm{d}}{\mathrm{d}s}(T \cdot B) &= \kappa(N \cdot B) - \tau(T \cdot N) \\
\frac{\mathrm{d}}{\mathrm{d}s}(N \cdot B) &= -\kappa(T \cdot B) + \tau(B \cdot B) - \tau(N \cdot N) \\
\frac{\mathrm{d}}{\mathrm{d}s}(T \cdot T) &= 2\kappa(T \cdot N) \\
\frac{\mathrm{d}}{\mathrm{d}s}(N \cdot N) &= -2\kappa(T \cdot N) + 2\tau(N \cdot B) \\
\frac{\mathrm{d}}{\mathrm{d}s}(B \cdot B) &= -2\tau(N \cdot B).
\end{aligned}$$

These are a system of ODEs for the quantities $T \cdot N, T \cdot B$, etc with initial conditions $0, 0, 0, 1, 1, 1$. However, the system can also be solved by $T \cdot N \equiv 0$, $T \cdot B \equiv 0$, etc, and so by uniqueness of solutions to linear ODEs, it follows that $T \cdot N = 0$, etc, which proves the orthonormality. To show positive orientation, it suffices to show that $(T \times N) \cdot B \equiv 1$. This is true at the basepoint of time by choice of initial conditions, and if we take the derivative, we find

$$\frac{\mathrm{d}}{\mathrm{d}s}((T \times N) \cdot B) = \kappa(N \times N) \cdot B + [(T \times (-\kappa T)) + T \times (\tau B)] \cdot B + (T \times N) \cdot (-\tau N),$$

which we see to be equal to zero by our orthonormality proof from above. Thus, $\{T, N, B\}$ is indeed a positively-oriented orthonormal basis.

Finally, we need to show that there exists a unique curve with T as its unit tangent (from which the remainder of the quantities N, etc will follow); indeed, we have

$$\gamma : I \to \mathbb{R}^3, \quad \gamma(s) = p_0 + \int_{s_0}^{s} T(u)\,\mathrm{d}u$$

is the unique curve with $\dot{\gamma} = T$; the fact that $\gamma \in C^k$ follows from $T \in C^{k-1}$. ■

Exercise 2.17: With the same assumptions as above, also assume $\sigma : I \to \mathbb{R}_{>0} \in C^{k-1}$. Then, there exists a unique C^k regular path $\gamma \in \mathbb{E}^3$ such that

$$\|\dot{\gamma}\| = \sigma, \quad \kappa_\gamma = \kappa, \quad \tau_\gamma = \tau.$$

We're interested in defining the torsion for more general paths in a consistent way. Let γ a regular C^3 curve in $\mathbb{R}^3$ with $\kappa > 0$. Let $\tilde{\gamma} : \tilde{I} \to \mathbb{R}^3$ be a arc-length reparametrization using $t : \tilde{I} \to I$, and let $s = t^{-1}$, and define

$$\tau := \tilde{\tau} \circ s,$$

where $\tilde{\tau}$ the torsion of $\tilde{\gamma}$, as defined above.

↪**Proposition 2.3**: Let γ be as above. Then,

$$\kappa = \frac{\|\dot{\gamma} \times \ddot{\gamma}\|}{\|\dot{\gamma}\|^3}, \quad \tau = \left(\frac{\dot{\gamma} \times \ddot{\gamma}}{\|\dot{\gamma} \times \ddot{\gamma}\|^2}\right) \cdot \dddot{\gamma}$$

Proof. We know $\kappa = \frac{\|\ddot{\gamma}^\perp\|}{\|\dot{\gamma}\|^2}$. In $\mathbb{R}^3$, $\|\dot{\gamma} \times \ddot{\gamma}\|$ is the area of the parallelogram with sides $\dot{\gamma}, \ddot{\gamma}$, or equivalently, twice the area of the triangle with base $\dot{\gamma}$ and height $\ddot{\gamma}^\perp$ (the perpendicular to the base $\dot{\gamma}$), i.e.

$$\|\dot{\gamma} \times \ddot{\gamma}\| = \|\dot{\gamma}\|\|\ddot{\gamma}^\perp\|,$$

which proves the first claim. The second claim follows from lots of careful chain rules. ■

Exercise 2.18: Is torsion preserved by reversals? i.e., if $\overline{\gamma} := \gamma \circ \overline{t}$ where $\overline{t}(t) = -t$, is $\tau_{\overline{\gamma}}(\overline{t}) = \tau_\gamma(t)$?

Solution. Given a curve $\gamma : I \to \mathbb{R}^3$, with $\overline{\gamma} := \gamma \circ \overline{t}$, note that $\dot{\overline{t}} = -1$ and all the higher derivatives are zero. Thus, we can easily compute

$$\dot{\overline{\gamma}} = -\dot{\gamma} \circ \overline{t},$$
$$\ddot{\overline{\gamma}} = \ddot{\gamma} \circ \overline{t},$$
$$\dddot{\overline{\gamma}} = -\dddot{\gamma} \circ \overline{t},$$

so that

$$(\dot{\overline{\gamma}} \times \ddot{\overline{\gamma}}) \cdot \dddot{\overline{\gamma}} = (\dot{\gamma} \circ \overline{t} \times \ddot{\gamma} \circ \overline{t}) \cdot \dddot{\gamma} \circ \overline{t}, \quad \|\dot{\overline{\gamma}} \times \ddot{\overline{\gamma}}\| = \|\dot{\gamma} \circ \overline{t} \times \ddot{\gamma} \circ \overline{t}\|$$

from which the claim easily follows. ■

Exercise 2.19: (Twisted Cubic) Let $\gamma : \mathbb{R} \to \mathbb{R}^3$ be given by $\gamma(t) = \left(t, t^2, t^3\right)$. Show that $\kappa(0) = 2, \tau(0) = 3$.

Exercise 2.20: (Helix) Let $\gamma(t) = (\cos(t), \sin(t), t)$. Show that $\kappa \equiv \frac{1}{2}, \tau \equiv \frac{1}{2}$.

Exercise 2.21: Find an example where $\kappa \equiv \frac{1}{2}, \tau \equiv -\frac{1}{2}$.

§2.5 Global Theorems/Properties of Plane Curves

Let $\mathbb{S}^1$ denote the unit circle in $\mathbb{R}^2$ centered at the origin, with global periodic parametrization $\rho(t) = (\cos(t), \sin(t))$. Given a C^0 curve in $\mathbb{S}^1$ by $g : I \to \mathbb{S}^1$, a function $\theta : I \to \mathbb{R}$ is called a *lift* of g via ρ if

1. it is C^0
2. $g = \rho \circ \theta$

↪**Theorem 2.7**: Fix $t_0 \in \mathbb{R}, \theta_0 \in \mathbb{R}$ such that $g(t_0) = (\cos\theta_0, \sin\theta_0)$. Then, there exists a unique lift θ of g such that $\theta(t_0) = \theta_0$.

If $g : \mathbb{R} \to \mathbb{S}^1$ a periodic path with period $[a, b]$, then for any lift θ of g,

$$|\theta(b) - \theta(a)| = 2\pi n, \quad n \in \mathbb{Z}_+,$$

where n the number of times the curve "goes around" $\mathbb{S}^1$.

↪**Theorem 2.8** (Hopf's Umlanfasatz): If $C \subset \mathbb{R}^2$ a regular closed curve periodic (with period $[a, b]$) parametrization $\gamma : R \to \mathbb{R}^2$, then for any lift θ of its tangent vector T (i.e., θ is an angle function), $|\theta(b) - \theta(a)| = 2\pi$.

We say γ is *positively/ccw oriented* if $\theta(b) - \theta(a) = 2\pi$, and *negatively/cw oriented* if $\theta(b) - \theta(a) = -2\pi$.

↪**Theorem 2.9** (Jordan Curve Theorem): Let $C \subset \mathbb{R}^2$ a regular closed curve. Then, $\mathbb{R}^2 \setminus C$ has two connected components; one bounded ("inside" of C) and one unbounded ("outside" of C).

We can then say γ is *positively oriented* if T^* points inside C, and *negatively oriented* if T^* points outside C. It turns out these different notions of orientation are equivalent.

↪**Theorem 2.10** (Isoperimetric Inequality): Let $C \subset \mathbb{R}^2$ a regular closed curve. Let ℓ = length of C and A = area of inside of C. Then,

$$4\pi A \leq \ell^2,$$

with equality iff C is a circle.

Proof. (Sketch of Hopf's) ■

§3 Surfaces

§3.1 Preliminary Definitions

Throughout, let $n \geq 2$ and k an integer between 1 and ∞ (possibly infinity).

↪**Definition 3.1** (Surfaces): A subset $S \subset \mathbb{R}^n$ is a *regular C^k surface* if for all $p \in S$, there exists an open subset $U \subset \mathbb{R}^2$, a neighborhood $V \subset \mathbb{R}^n$ containing p, and a C^k function $\varphi : U \to \mathbb{R}^n$ such that

1. $\varphi(U) = V \cap S$, $\varphi : U \to V \cap S$ a homeomorphism (continuous bijection with continuous inverse)
2. $\mathrm{d}\varphi_q : \mathbb{R}^2 \to \mathbb{R}^n$ is one-to-one for all $q \in \mathcal{U}$.

Remark 3.1:

2. is equivalent to $J_\varphi(q) \in \mathbb{R}^{n\times 2}$ having rank 2, i.e. linearly independent columns.

↪**Theorem 3.1** (Inverse Function Theorem): Suppose $\tilde{U} \subset \mathbb{R}^m$ open and $f : \tilde{U} \to \mathbb{R}^m$ is C^k and $q \in \tilde{U}$ is such that $\mathrm{d}f_q : \mathbb{R}^m \to \mathbb{R}^m$ is invertible. Then there exist open sets $U \subset \tilde{U}$ and $V \subset \mathbb{R}^m$ with $q \in U$, and such that $f(U) = V, f|_U : U \to V$ is one-to-one with C^k inverse $g : V \to U$, and moreover,

$$J_g(f(q)) = J_f^{-1}(q).$$

↪**Proposition 3.1** (Graphs): If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}$ is a C^k function, then the graph of f, $\mathrm{graph}(f) := \{(x,y,z) \in \mathbb{R}^3 : f(x,y) = z\}$ is a regular C^k surface.

Proof. Let $V = \mathbb{R}^3$ and $\varphi : U \to \mathbb{R}^3$ be given by

$$\varphi(x,y) = (x,y,f(x,y)),$$

then $S := \varphi(U) = \mathrm{graph}(f)$. Then, note that φ is clearly injective (suffices to look at the first two coordinate functions), and by constructive surjective onto S. It is also continuous since the first two coordinates are linear and the last is C^k. We need to check the inverse; define $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ by $\pi(x,y,z) = (x,y)$. Then,

$$\pi|_S \circ \varphi : U \to U = \mathrm{id},$$

hence $\pi|_S$ is the inverse of φ. Its clear that $\pi|_S$ is continuous and thus φ indeed a homeomorphism.

Next we need to check the Jacobian of φ. We easily compute

$$J_\varphi(x,y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \partial_x f(x,y) & \partial_y f(x,y) \end{pmatrix}$$

which is of rank 2. ■

↪**Definition 3.2** (Critical points/Regular values): Let $U \subset \mathbb{R}^m$ open and $F : U \to \mathbb{R}^n$ be C^1. Then a point $p \in U$ is said to be a *critical point* of F if $\mathrm{d}F_p : \mathbb{R}^m \to \mathbb{R}^n$ is *not* onto (i.e., $J_F(p)$ has rank $< n$).

A point $r \in \mathbb{R}^n$ is called a *regular value* if it is *not* the image of a critical point, i.e.

$$\text{regular values} = \mathbb{R}^n \setminus F(\{\text{critical points}\}).$$

Remark 3.2: If $n = 1$, then "$\mathrm{d}F_p$ not onto" $\Leftrightarrow J_F(p) = \nabla F(p)^T = 0$.

↪**Theorem 3.2** (Level Sets): Suppose $\tilde{V} \subset \mathbb{R}^3$ open and $F : \tilde{V} \to \mathbb{R}$ a C^k function, and $a \in F(\tilde{V})$ a regular value. Then, $F^{-1}(a) = \{(x,y,z) \in \tilde{V} : F(x,y,z) = a\}$ is a regular C^k surface.

Proof. Let $p \in F^{-1}(a)$ (so $F(p) = a$). Since p a regular value, $\nabla F(p) \neq 0$, so in particular at least one coordinate of the gradient is nonzero, say (wlog) $\frac{\partial F}{\partial z}(p) \neq 0$. Define the function

$$G : \tilde{V} \to \mathbb{R}^3, \quad G(x,y,z) := (x,y,F(x,y,z)).$$

Define $\tilde{S} := G(S) = \{(x,y,a)\}$. Then, we check

$$J_G(p) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ - & \nabla F(p)^T & - \end{pmatrix}$$

which has determinant $\det J_G(p) = \frac{\partial F}{\partial z}(p) \neq 0$ i.e. is invertible. Since we also see G is C^k, we can appeal to the inverse function theorem. This means we have an open set $\subset \tilde{V}$ with $p \in V$, an open set $\tilde{U} \subset \mathbb{R}^3$ with $G(p) \in \tilde{U}$ with $G(V) = \tilde{U}$ and such that G a C^k homeomorphism on V, and a local inverse $H : \tilde{U} \to V$ which is also a C^k homeomorphism. Let $U = \tilde{U} \cap \tilde{S}$, viewed as living on the x-y plane (formally, write $\psi : U \to \tilde{U} \cap \tilde{S}$ with $\psi(x,y) = (x,y,a)$, obviously a smooth homeomorphism). Set now $\varphi = H(\circ\, \psi) : U \to F^{-1}(a) \cap V$. By chain rule, $J_\varphi(q) = J_H(\psi(q))J_\psi(q)$, which one checks is rank 2 for $q \in U$. ■

⊛ **Example 3.1**:

Remark 3.3: Everything we've said about surfaces in $\mathbb{R}^3$ apply similarly to "curves" in $\mathbb{R}^2$ (allowing curves without endpoints and with possibly multiple connected components).

More generally, for $1 \leq m \leq n$ and $1 \leq k \leq \infty$, we say $M \subset \mathbb{R}^n$ a regular C^k m-manifold if it satisfies the same definition as the above for surfaces, but replacing $\mathbb{R}^n$ for $\mathbb{R}^3$ and $\mathbb{R}^m$ for $\mathbb{R}^2$ and rank m for rank 2.

↪**Theorem 3.3** (Surfaces are local graphs): $S \subset \mathbb{R}^3$ is a regular C^k surface $\Leftrightarrow$ for all $p \in S$, there exists an open set $U \subset \mathbb{R}^2$ and $V \subset \mathbb{R}^3$ with $p \in V$ such that

$$V \cap S = \text{the graph of some } C^k \text{ function } U \to \mathbb{R}.$$

In other words, S is always locally given by $z = f(x,y)$ *or* $y = g(x,z)$ *or* $x = h(y,z)$.

Proof. Use the inverse function theorem. ■

⊛ **Example 3.2**:

↪**Proposition 3.2**: Suppose $S \subset \mathbb{R}^3$ is a regular C^k surface and $U_1, U_2 \subset \mathbb{R}^2$ are two open sets with two C^k parametrizations $\varphi_i : U_i \to S$ with rank 2 Jacobians on U_i. Then, $g = \varphi_2^{-1} \circ \varphi_1 : U_1 \to U_2$ and $h = \varphi_1^{-1} \circ \varphi_2 : U_2 \to U_1$ are C^k homeomorphisms. Thus, we can write $\varphi_2 = \varphi_1 \circ h$, and we call φ_2 a *reparametrization* of φ_1 using change of parameters h.

Proof. (Sketch) It suffices to prove it for h. Let $q_2 \in U_2$ and $q_1 = h(q_2)$. Change coordinates in $\mathbb{R}^3$ so that $\mathrm{d}\varphi_1(\mathbb{R}^2)$ is the (x_1, x_2)-plane. In particular this means that the first two rows of $J_{\varphi_1}(q_1)$ are linearly independent. Let $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ be the projection onto the first two coordinates. Let $\Phi_1 := \pi \circ \varphi_1 : U_1 \to \mathbb{R}^2$, which is clearly C^k. Compute that $J_{\Phi_1}(q_1)$ is invertible, hence we can apply the inverse-function theorem to obtain $U_1' \subset U_1$ and $V_1 \subset \mathbb{R}^2$ with $q_1 \in U_1'$ and with $\Phi_1(U_1') = V_1$ and Φ_1 restricted to U_1' is a C^k bijection with C^k inverse, denoted ψ_1. Let $\Phi_2 := \pi \circ \varphi_2$ and $U_2' := \Phi_2^{-1}(V_1')$. Then, one can check that $\psi_1 \circ \Phi_2$ is a restriction of h to U_2', and since ψ_1 (by IFT) C^k and Φ_2 C^k, this implies h locally C^k. Since q_2 arbitrary, this means $h \in C^k(U_2)$ indeed. ■

↪**Definition 3.3** (Surface Patch): Given an open set $U \subset \mathbb{R}^2$ open, we say a function $\varphi : U \to \mathbb{R}^3$ is (regular C^k) *surface patch* if φ is C^k and $\mathrm{d}\varphi_q : \mathbb{R}^2 \to \mathbb{R}^3$ is one-to-one for all $q \in U$.

Remark 3.4: φ needn't be one-to-one, unlike in the definition of a surface.

Remark 3.5: There is no real analog to arc-length parametrization in the context of surfaces, and hence no "canonical" parametrization as for curves. Hence, we will need to be more careful in what follows to properly define quantities for surfaces to be parametrization-invariant where necessary.

↪**Definition 3.4** (Tangent Space): Fix $q \in U$. The *tangent space*

$$T_q\varphi := \mathrm{d}\varphi_q(\mathbb{R}^2)$$

to a surface patch φ at $q \in U$ is the image of $\mathrm{d}\varphi_q$ in $\mathbb{R}^3$, i.e. a two-dimensional vector space spanned by the columns of the Jacobian $J_\varphi(q)$.

↪**Proposition 3.3**: If $\tilde{\varphi} : \tilde{U} \to \mathbb{R}^3$ a reparametrization of $\varphi : U \to \mathbb{R}^3$ using $h : \tilde{U} \to U$, then

$$T_{\tilde{q}}\tilde{\varphi} = T_q\varphi, \quad q := h(\tilde{p}).$$

Remark 3.6: Notably, the Jacobians $J_\varphi, J_{\tilde{\varphi}}$ may look different, but this proposition says that they have the same image.

Proof. By chain rule,

$$\mathrm{d}\tilde{\varphi}_{q'} = \mathrm{d}\varphi_{h(q')} \circ \mathrm{d}h_{q'} = \mathrm{d}\varphi_q \circ \mathrm{d}h_{q'}$$

thus

$$T_{q'}\tilde{\varphi} = \mathrm{d}\tilde{\varphi}_{q'}(\mathbb{R}^2) = \mathrm{d}\varphi_q \circ \mathrm{d}h_{q'}(\mathbb{R}^2) = \mathrm{d}\varphi_q(\mathbb{R}^2) = T_q\varphi,$$

where we use the fact that $\mathrm{d}h_{q'} : \mathbb{R}^2 \to \mathbb{R}^2$ is invertible so in particular surjective. ■

↪**Definition 3.5**: Let $S \subset \mathbb{R}^3$ be a regular C^k surface and let $p \in S$. Define the *tangent space* to S at p by

$$T_pS = T_q\varphi$$

where $\varphi : U \to \mathbb{R}^3$ any local parametrization of S near p with $q \in U$ such that $\varphi(q) = p$.

↪**Corollary 3.1**: This is a well-defined object, i.e. independent of choice of parametrization φ.

Proof. This follows directly from the previous proposition. ■

Exercise 3.1:

1. Similarly define $T_p\mathcal{C}$ for a regular C^k curve $\mathcal{C}$ with $p \in \mathcal{C}$, and moreover show that this is well-defined.
2. Show that if $\mathcal{C}$ a curve contained in a surface $\mathcal{S}$, then for any $p \in \mathcal{C}$, we have that $T_p\mathcal{C} \subset T_pS$.

⊛ **Example 3.3**: Let $\mathbb{S}^2$ be the unit sphere centered at 0 and take $p = \frac{1}{\sqrt{3}}(1,1,1) \in \mathbb{S}^2$. Let

$$\varphi_N : U_N \to \mathbb{R}^3, \quad \varphi_N(x,y) = \left(x, y, \sqrt{1-x^2-y^2}\right), \quad q := \frac{1}{\sqrt{3}}(1,1) \in U_N := \{x^2+y^2<1\}.$$

Note then that $\varphi_N(q) = p$, and

$$\varphi_N'(q) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{x}{\sqrt{1-x^2-y^2}} & -\frac{y}{\sqrt{1-x^2-y^2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$$

thus

$$T_q\varphi_N = \operatorname{span}\{(1,0,-1),(0,1,-1)\}.$$

Taking $\varphi_E(x,z) := \left(x, \sqrt{1-x^2-z^2}, z\right) : U_E := \{x^2+z^2<1\} \to \mathbb{R}^3$ and $q' = \frac{1}{\sqrt{3}}(1,1)$, one similarly finds

$$\varphi_E'(q') = \begin{pmatrix} 1 & 0 \\ -1 & -1 \\ 0 & 1 \end{pmatrix} \Rightarrow T_{q'}\varphi_E = \operatorname{span}\{(1,-1,0),(0,-1,1)\}.$$

↪**Definition 3.6** (Unit Normal): A vector $n \in \mathbb{R}^3$ is called the *unit normal* to a regular surface S at a point p if $\|n\| = 1$ and n is orthogonal to T_pS.

Remark 3.7: Since T_pS a 2-dimensional subspace, there are exactly two unit normals to S at p, which are negatives of each other.

↪**Definition 3.7** (Orientation): An *orientation* on a surface S is a continuous map $N : S \to \mathbb{S}^2$ such that $N(p)$ is a unit normal to S at p for all $p \in S$. We say then S *orientable* if such a function exists on S.

Remark 3.8: If S connected and orientable, there exist exactly two orientations on S (why?).

↪**Proposition 3.4** (Level sets are orientable): Let $\tilde{V} \subset \mathbb{R}^3$ open and $F : \tilde{V} \to \mathbb{R}$ a C^1 function. Then for any $a \in F(\tilde{V})$ a regular value for F, $S := F^{-1}(a)$ is orientable.

Proof. For $p \in S$, $\nabla F(p) \neq 0$ since p is not a critical point. Define $N : S \to \mathbb{R}^3$ by

$$N(p) := \frac{\nabla F(p)}{\|\nabla F(p)\|}.$$

This is clearly continuous since $F \in C^1$, and well-defined since ∇F doesn't vanish over S. Moreover, if $\varphi : U \to S$ a local parametrization of S near p, then for $x \in U$,

$$a = F(\varphi(x)) \Rightarrow 0 = \nabla[F(\varphi(x))] = \nabla F(\varphi(x)) \cdot \mathrm{d}\varphi(x),$$

which implies ∇F at p is orthogonal to the columns of $\mathrm{d}\varphi_p$ and hence the tangent space of S. ■

Exercise 3.2: Show that graphs are orientable.

⊛ **Example 3.4** (The Mobius strip: a non-orientable surface): Let

$$\gamma : \mathbb{R} \to \mathbb{R}^3, \quad t \mapsto (2\cos t, 2\sin t, 0),$$

and define

$$\varphi : (-1,1) \times \mathbb{R} \to \mathbb{R}^3, \quad \varphi(s,t) := \gamma(t) + s\left[\cos\left(\frac{t}{2}\right)N(t) + \sin\left(\frac{t}{2}\right)B(t)\right].$$

We claim that $S := \varphi((-1,1) \times \mathbb{R})$ a non-orientable, regular surface. We leave regularity as an exercise. For non-orientability, we find that

$$J_\varphi(0,t) = \begin{pmatrix} -\cos(t)\cos\left(\frac{t}{2}\right) & -2\sin(t) \\ -\sin(t)\cos\left(\frac{t}{2}\right) & 2\cos(t) \\ \sin\left(\frac{t}{2}\right) & 0 \end{pmatrix}.$$

Any normal vector at t must be orthogonal to the columns of this Jacobian, i.e. a multiple of

$$\mathrm{d}\varphi_{0,t}(e_1) \times \mathrm{d}\varphi_{0,t}(e_2) = \left(-2\cos(t)\sin\left(\frac{t}{2}\right), -2\sin(t)\sin\left(\frac{t}{2}\right), -2\cos\left(\frac{t}{2}\right)\right).$$

This converges to $(0,0,-2)$ as $t \to 0$ and $(0,0,2)$ as $t \to 2\pi$. But our surface parametrization is 2π-periodic in t, so these vectors should be equal, but aren't hence there can't exists a continuous map in t to a normal vector can't be continuous.

§3.2 Functions on Surfaces and the First Fundamental Form

↪**Definition 3.8**: Let $S \subset \mathbb{R}^3$ a regular C^k surface. A function $f : S \to \mathbb{R}^n$ is C^k if for all p, there exists a local parametrization $\varphi : U \to S$ near p such that $f \circ \varphi : U \to \mathbb{R}^n$ is C^k; we call the expression $f \circ \varphi$ to be f *in local coordinates*.

↪**Lemma 3.1**: $f : \tilde{S} \to S$ is C^k iff for all $\tilde{p} \in \tilde{S}$ there are local parametrizations $\tilde{\varphi} : \tilde{U} \to \tilde{S}$ near $\tilde{p}$ and $\varphi : U \to S$ near $f(\tilde{p})$ such that $\varphi^{-1} \circ f \circ \tilde{\varphi} : \tilde{U} \to U$ is C^k.

Exercise 3.3: Let $f : \tilde{S} \to S$ be C^1, then $\mathrm{d}f_{\tilde{p}} : T_{\tilde{p}}\tilde{S} \to T_{f(\tilde{p})}S$ is well-defined.

Exercise 3.4: Define $\pi : \mathbb{R}^2 \to \mathbb{S}^2$ be the stereographic parametrization of the sphere based on the north pole $N = (0,0,1)$. Fix $c \in \mathbb{R}^2$ and let $M_c : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $M_c(x) := x + c$. Show that

$$f_c : \mathbb{S}^2 \to \mathbb{S}^2, \quad f_c(p) := \begin{cases} N & p = N \\ (\pi \circ M_c \circ \pi^{-1})(p) & p \neq N \end{cases}$$

is smooth.

↪**Definition 3.9** (1st Fundamental Form): Let $S \subset \mathbb{R}^3$ a regular surface. The *first fundamental form* assigns to each $p \in S$ the quadratic form

$$\|\cdot\|^2 : T_p(S) \to \mathbb{R}, \quad v \mapsto \|v\|^2.$$

↪**Definition 3.10** (Isometry): A C^1 function $f : \tilde{S} \to S$ is a *local isometry* if for all $\tilde{v} \in T_{\tilde{p}}\tilde{S}$,

$$\|\tilde{v}\|^2 = \left\|\mathrm{d}f_{\tilde{p}}(\tilde{v})\right\|^2.$$

We say f an *isometry* if it is a local isometry and a bijection.

Exercise 3.5: If $f : \tilde{S} \to S$ a restriction of a rigid motion of $\mathbb{R}^3$, then it is an isometry.

↪**Definition 3.11**: Let $\varphi : U \to S$ be a local parametrization of a regular surface S near p. Define for $q \in U$,

$$I_q : \mathbb{R}^2 \to \mathbb{R}, \quad u \mapsto \left\|\mathrm{d}\varphi_q(u)\right\|^2$$

and

$$\langle u_1, u_2 \rangle_q := \frac{1}{4}\big[I_q(u_1 + u_2) - I_1(u_1 - u_2)\big].$$

Exercise 3.6: The map $(u_1, u_2) \mapsto \langle u_1, u_2 \rangle_q$ defines an inner product on $\mathbb{R}^2$.

Remark that $\langle u_1, u_2 \rangle_q = \mathrm{d}\varphi_q(v) \cdot \mathrm{d}\varphi_q(v')$. Moreover, if we pick local coordinates for$\mathbb{R}^3$, we find

$$\langle v, v' \rangle_q = \big(J_\varphi(q)v\big) \cdot \big(J_\varphi(q)v'\big) = v^T J_\varphi^T(q) J_\varphi(q) v',$$

where we notice $J_\varphi^T(q)J_\varphi(q)$ a symmetric 2×2 matrix. We conventionally write

$$J_\varphi^T(q)J_\varphi(q) = \begin{pmatrix} E(q) & F(q) \\ F(q) & G(q) \end{pmatrix}, \quad E(q) = \|\partial_u \varphi\|^2, G(q) = \|\partial_v \varphi\|^2, F(q) = \partial_u \varphi \cdot \partial_v \varphi$$

where we write (u, v) for coordinates in $\mathbb{R}^2$ (and all the expressions above are evaluated at some point q). Thus if $v = (v_1, v_2) \in \mathbb{R}^2$, we find

$$\mathrm{I}_q(v) = E(q)v_1^2 + 2F(q)v_1v_2 + G(q)v_2^2$$

which we call the first-fundamental form in local coordinates (since the expressions E, F, G depend on choice of coordinates). We write, more concisely,

$$\mathrm{I} = E\,\mathrm{d}u^2 + 2F\,\mathrm{d}u\,\mathrm{d}v + G\,\mathrm{d}v^2$$

again locally, where $\mathrm{d}u$, etc, shorthand for the function $\mathrm{d}u[(v_1, v_2)] := v_1$, etc.

⊛ **Example 3.5**: The cylinder $S_1 = \{x^2 + y^2 = 1\}$, parametrized by $\varphi_1(s,t) = (\cos(s), \sin(s), t)$ has first fundamental form

$$\mathrm{d}s^2 + \mathrm{d}t^2$$

is local coordinates ($E = G = 1, F = 0$). In coordinates $\psi : \mathbb{R}^2 - \{0\} \to \mathbb{R}^3$,

$$\psi(u,v) = \left(\frac{u}{\|(u,v)\|}, \frac{v}{\|(u,v)\|}, \frac{1}{2}\ln\left(u^2 + v^2\right) \right)$$

gives first fundamental form

$$\frac{\mathrm{d}u^2 + \mathrm{d}v^2}{u^2 + v^2}.$$

⊛ **Example 3.6** (Computing Length/Area in Local Coordinates):

⊛ **Example 3.7** (Conformal Parametrizations, Stereographic Parametrization):

§3.3 Curvature and the Second Fundamental Form

Given $S \subset \mathbb{R}^3$ an oriented regular C^2 surface with normal N, we can view N, since it is a unit vector, as a map $N : S \to \mathbb{S}^2$; this is C^1. In particular, N is given by the (plus or minus) of the cross product of the columns of the Jacobian of a local parametrization. We call this map the *Gauss map*.

↪**Definition 3.12** (Shape Operator): With $p \in S$ as above, the *shape operator* at p is the map

$$-\mathrm{d}N_p : T_pS \to T_{N(p)}\mathbb{S}^2.$$

The *Gaussian curavture* at p is

$$K(p) := \det\left(-\mathrm{d}N_p\right), \quad K : S \to \mathbb{R}$$

and the *mean curvature* at p is

$$H(p) := \frac{1}{2}\operatorname{tr}\left(-\mathrm{d}N_p\right), \quad H : S \to \mathbb{R}.$$

Remark 3.9: Note that $T_{N(p)}\mathbb{S}^2 = T_pS$, since both are planes normal to $N(p)$, which is unique.

⊛ **Example 3.8** (Plane):

⊛ **Example 3.9** (Sphere):

⊛ **Example 3.10** (Curvature at Critical Point of a Function Graph):

↪**Proposition 3.5**: $K(p)$ is invariant under change of orientation. $H(p)$ is reversed by change of orientation.

Remark 3.10: This means that $K(p)$ and $|H(p)|$ is well-defined, locally, for even non-orientable surfaces, since we can always locally define a (continuous) unit normal by taking the (normalized) cross product of the columns of a local parametrization.

↪**Proposition 3.6**: K, H are preserved by rigid motion. More precisely, let $S \subset \mathbb{R}^3$ a regular C^2 surface with orientation N and $M : \mathbb{R}^3 \to \mathbb{R}^3$ a rigid motion (where $M(x) = O(x) + b$ for O an orthogonal matrix and b a fixed vector). Then, $\tilde{S} = M(S)$ is a regular C^2 surface with orientation $\tilde{N}(\tilde{p}) := O(N(p))$, $\tilde{p} = M(p)$, so in particular,

$$\tilde{K}(\tilde{p}) = K(p), \quad \tilde{H}(\tilde{p}) = H(p).$$

Remark 3.11: Every point of a regular C^2 surface is "essentially" a critical point of a function, since we may always rotate the surface such that the tangent plane at a given point is the $x - y$ plane (and by the previous proposition, the curvatures won't be affected) and every regular surface is locally a graph.

↪**Corollary 3.2**: The shape operator is self-adjoint with respect to the standard inner product.

We provide two proofs, for the second gives useful insights for later.

Proof *1*. This follows directly from the previous remarks for we know that we can always pick coordinates so that $-\mathrm{d}N_p = H_f$, the Hessian of some C^2 function f. By Clairault's theorem, this is always symmetric and thus $-\mathrm{d}N_p$ is self-adjoint. ■

Proof *2*. Choose local parametrization $\varphi : U \to S$ near a point $p = \varphi(q) \in S$, so $\partial_u\varphi, \partial_v\varphi$ (evaluated at q) provide a basis for T_pS. Because these are a basis, it suffices to show by linearity that

$$-\mathrm{d}N_p(\partial_u\varphi) \cdot \partial_v\varphi = \partial_u\varphi \cdot \big(-\mathrm{d}N_p(\partial_v\varphi)\big),$$

for the other possible combinations we need to check (eg $\mathrm{d}N_p(\partial_u\varphi) \cdot \partial_u\varphi$, etc) are trivial. Since $N \circ \varphi$ is orthogonal to $\partial_u\varphi$ throughout U, this means

$$0 \equiv (N \circ \varphi) \cdot \partial_u\varphi$$

so taking the derivative with respect to v, we get

$$0 = (\partial_v(N \circ \varphi)) \cdot \partial_u\varphi + (N \circ \varphi) \cdot \partial^2_{vu}\varphi.$$

Similar occurs if we begin with orthogonality to $\partial_v\varphi$ and take a derivative with respect to v, i.e.

$$0 = (\partial_u(N \circ \varphi)) \cdot \partial_v\varphi + (N \circ \varphi) \cdot \partial^2_{uv}\varphi.$$

Since φ C^2, $\partial^2_{uv}\varphi = \partial^2_{vu}\varphi$ and so these two equations combined imply

$$-(\partial_u(N \circ \varphi)) \cdot (\partial_v\varphi) = (\partial_u\varphi) \cdot (-\partial_v(N \circ \varphi)).$$

But in local coordinates, $-\mathrm{d}N_p(\partial\varphi_u) = \partial_u N \circ \varphi$, etc., so the proof follows. ■

↪**Corollary 3.3**: There exists an orthonormal basis for T_pS such that $-\mathrm{d}N_p$ is represented by a diagonal matrix $k_1 \leq k_2$. We call k_1, k_2 the *principal curvatures* of S at p, and their corresponding eigenvectors the *principal directions* of S at p.

In this notation, $K(p) = k_1 k_2, H(p) = \frac{1}{2}(k_1 + k_2)$.

↪**Definition 3.13**: Let $p \in S$. We say that p is

- *elliptic* if $K(p) > 0$
- *hyperbolic* if $K(p) < 0$
- *planar* if $K(p) = 0$ but $H(p) \neq 0$
- *umbilic* if $k_1 = k_2$.

↪**Definition 3.14**: The *second fundamental form* of S at p is the map

$$\mathbb{I}_p : T_pS \to \mathbb{R}, \quad \mathbb{I}_p(v) = -\mathrm{d}N_p(v) \cdot v.$$

In local coordinates $\varphi : U \to S, p = \varphi(q)$, we associate $\mathbb{I}_q$ with the matrix

$$\begin{pmatrix} e & f \\ f & g \end{pmatrix} : \mathbb{R}^2 \to \mathbb{R}^2,$$

with respect to the standard basis of $\mathbb{R}^2$, or equivalently with the quadratic form

$$e\,\mathrm{d}u^2 + 2f\,\mathrm{d}u\,\mathrm{d}v + g\,\mathrm{d}v^2.$$

Note that

$$\begin{aligned} e &= -\mathrm{d}N_p(\partial_u\varphi) \cdot \partial_u\varphi = (N \circ \varphi) \cdot \partial^2_{uu}\varphi \\ f &= -\mathrm{d}N_p(\partial_u\varphi) \cdot \partial_v\varphi = -\mathrm{d}N_p(\partial_v\varphi) \cdot \partial_u\varphi = (N \circ \varphi) \cdot \partial^2_{uv}\varphi \\ g &= -\mathrm{d}N_p(\partial_v\varphi) \cdot \partial_v\varphi = (N \circ \varphi) \cdot \partial^2_{vv}\varphi, \end{aligned}$$

as discussed in the proof of the self-adjointness of $-\mathrm{d}N_p$ (all evaluated at q).

Remark 3.12: The advantage of the latter formulae is that one needn't compute the derivatives of N in local coordinates, which isn't usually the easiest.

Remark 3.13: To be clear, given $v \in T_pS$, one can write $v = \mathrm{d}_q\varphi(u)$ for some $u = (u_1, u_2) \in \mathbb{R}^2$. Then,

$$\mathbb{I}_p(v) = u^T \begin{pmatrix} e & f \\ f & g \end{pmatrix} u = eu_1^2 + 2fu_1u_2 + gu_2^2.$$

↪**Proposition 3.7** (Shape Operator in Local Coordinates): In local coordinates given by φ, the shaper operator $-\mathrm{d}N_p$ is represented by, with respect to the basis $\{\partial_u\varphi, \partial_v\varphi\}$, by

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \cdot \begin{pmatrix} e & f \\ f & g \end{pmatrix} = \frac{1}{EG - F^2} \begin{pmatrix} eG - fG & fG - gF \\ fE - eF & gE - fF \end{pmatrix}.$$

↪**Corollary 3.4** (Curvature in Local Coordinates):

$$K \circ \varphi = \frac{eg - f^2}{EG - F^2}, \quad H \circ \varphi = \frac{1}{2}\frac{eG - 2fF + gE}{EG - F^2}$$

↪**Definition 3.15** (Normal Curvature): The *normal curvatures* of a surface S at a point p along $v \in T_pS$ is $\mathbb{I}_p(v)$ where v a unit vector; the principal curvatures are the min, max normal curvatures on T_pS.

↪**Definition 3.16** (Normal Section): Let S a regular C^2 surface with orientation N. Fix $p \in S$ and $v \in T_pS$ and let $\mathcal{P}$ be the plane in $\mathbb{R}^3$ containing the point p and parallel to $N(p)$ and v (i.e. with normal $N(p) \times v$). The intersection $\mathcal{C} := \mathcal{P} \cap S$ is a regular C^2 curve near p; this is called the *normal section* to S at p along v.

↪**Proposition 3.8** (Normal Curvature Characterization via Curves): Let $v \in T_pS$ be a unit vector and $\varphi.(-\varepsilon, \varepsilon) \to S$ a regular C^2 path with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$. Then,

1. $\mathbb{I}_p(v) = N(p) \cdot \ddot{\gamma}(0)$
2. $\left|\mathbb{I}_p(v)\right| \leq K_\gamma(0)$, the curvature of the curve γ, with equality achieved along a normal section along v.

Proof. The inequality in 2. follows from Cauchy-Schwarz. We'll prove the "equality achieved" statement later. For 1., note that $\gamma \in S$ means $\dot{\gamma}(t) \in T_{\gamma(t)}S$ for all t, and thus $\dot{\gamma} \cdot N \circ \gamma \equiv 0$ for all t. Taking derivatives and evaluating at $t = 0$, we get

$$\begin{aligned} 0 &= \ddot{\gamma}(0) \cdot (N \circ (\gamma(0))) + \dot{\gamma}(0) \cdot \left(\mathrm{d}N_{\gamma(0)} \cdot \dot{\gamma}(0)\right) \\ &= \ddot{\gamma}(0) \cdot N(p) + v \cdot \mathrm{d}N_p(v) \\ &= \ddot{\gamma}(0) \cdot N(p) - \mathbb{I}_p(v), \end{aligned}$$

from which the proof follows.

Finally, if γ is a normal section with unit-speed parametrization along v, then $\ddot{\gamma}(0)$ is orthogonal to v and parallel to the plane $\mathcal{P}$ containing p and parallel to both $N(p)$ and v. Thus, it must be that $\dot{\gamma}(0)$ is parallel to $N(p)$, so equality holds in the Cauchy-Schwarz inequality in 1., and we find

$$\mathbb{I}_p(v) = \|N(p)\|\|\ddot{\gamma}(0)\| = \pm K_\gamma(0),$$

from which the proof follows by taking absolute values. ■

3.3.1 Orienting the Tangent Space

Let S a C^2 oriented surface and for $p \in S$, choose an orthonormal basis $\{v_1, v_2\}$ for T_pS with $v_1 \times v_2 = N$. This means $\{v_1, v_2, N\}$ forms a positively oriented orthonormal basis of $\mathbb{R}^3$.

Then, we can define "clockwise/counter-clockwise" directions for vectors in T_pS. Namely, given $w \in T_pS$, we can write $w = a_1v_1 + a_2v_2$ for some scalars a_1, a_2. Define then the $\frac{\pi}{2}$-counter-clockwise rotation of w by

$$w^* = av_2 - bv_1,$$

which one notices is still contained in T_pS.

Now, let $\gamma : I \to S$ be a unit-speed parametrized, C^2 path with $\gamma(0) = p$. Then, $\ddot{\gamma}(0)$ is orthogonal to $\dot{\gamma}(0)$ and thus $\ddot{\gamma}(0)$ lies in the plane normal to $\dot{\gamma}(0)$, which is spanned by $N(p)$ and $\dot{\gamma}(0)^*$. So, we can write, for some constants K_N, K_g,

$$\ddot{\gamma}(0) = K_N \cdot N(p) + K_g \cdot \dot{\gamma}(0)^*.$$

We call K_g the *geodesic curvature* and K_N the *normal curvature* of γ at 0. Remark that

$$K_N = \ddot{\gamma}(0) \cdot N(p) = \mathbb{I}_p(\dot{\gamma}(0)),$$

by the previous proposition, and

$$K_g = \ddot{\gamma}(0) \cdot \dot{\gamma}(0)^*,$$

which is the same as the formula for the signed curvature if γ were a planar path. Moreover,

$$K_\gamma(0)^2 = K_N^2 + K_g^2.$$

For more general non-unit speed paths γ, we define these quantities similarly in terms of their unit-speed reparametrization.

↪**Definition 3.17**: We call $\gamma : I \to S$ a regular C^2 path a *geodesic* if $\ddot{\gamma}$ is orthogonal to S everywhere in I.

↪**Proposition 3.9**: γ is a geodesic iff γ has constant speed and $K_g \equiv 0$.

↪**Proposition 3.10**:

§3.4 An Aside: the Fary-Milnor Theorem

↪**Definition 3.18** (Unknotted): Let $\mathcal{C} \subset \mathbb{R}^3$ a closed regular curve. We call $\mathcal{C}$ *unknotted* if there exists a subset $D \subset \mathbb{R}^3$ and a homeomorphism $h : \overline{\mathbb{D}} \to D$ (where $\mathbb{D}$ the unit disk) such that $\mathcal{C} = h(\mathbb{S}^1)$. We call $\mathcal{C}$ knotted if it is not unknotted.

↪**Theorem 3.4** (Fary-Milnor): Let $\mathcal{C}$ a closed, regular, C^4 curve. If $\mathcal{C}$ is knotted and $\kappa_{\mathcal{C}} > 0$ then

$$\int_{\mathcal{C}} \kappa_{\mathcal{C}} \, \mathrm{d}s \geq 4\pi.$$

We sketch the main ideas of the proof. Fix $\gamma : \mathbb{R} \to \mathcal{C}$ an arc-length, ℓ-periodic parametrization of $\mathcal{C}$. We need the following exercise:

Exercise 3.7: Let $\mathcal{C} \subset \mathbb{R}^3$ a regular C^3 closed curve with strictly positive curvature. Let, for $r > 0$ and $p \in \mathcal{C}$, the set $\mathcal{C}_{r,p}$ be the r-radius circle centered at p lying in the normal plane of $\mathcal{C}$ at p. Then, there exists $r > 0$ sufficiently small such that the set

$$\mathcal{C}_r := \bigcup_{r \in \mathcal{C}} \mathcal{C}_{r,p}$$

is a regular parametrized surface; we call this the r-tubular neighborhood of $\mathcal{C}$. Moreover, one can take $r < \frac{1}{\max \kappa_{\text{cal}}}$.
Hint: find a parametrization of $\mathcal{C}_{r,p}$ in terms of the unit normal, binormal of $\mathcal{C}$ at p, and extend this to a parametrization of $\mathcal{C}_r$. Differentiate this expression to check regularity.

Let $0 < \varepsilon < \frac{1}{\max \kappa_{\mathcal{C}}}$, an S be the *ε-tubular neighborhood of S* which has a (global, regular, C^2 parametrization)

$$\varphi : [0, \ell] \times [0, 2\pi] \to S, \quad \varphi(s, \theta) = \gamma(s) + \varepsilon \cos(\theta) n(s) + \varepsilon \sin(\theta) b(s)$$

where we write n, b for the unit normal, binormal resp. of $\mathcal{C}$ to not confuse with our notation N for the normal of a surface.

One computes

$$\|\partial_s \varphi \times \partial_\theta \varphi\| = \varepsilon\Big(1 - \varepsilon \cos(\theta) \kappa_\gamma(s)\Big), \quad K \circ \varphi = \frac{-\cos(\theta)\kappa_\gamma(s)}{\varepsilon\Big(1 - \varepsilon \cos(\theta)\kappa_\gamma(s)\Big)}.$$

Based on this expression for the Gaussian curvature of S, we see that the subset of S with positive curvature is precisely

$$S_+ := \{p \in S : K(p) \geq 0\} = \varphi\left([0, \ell] \times \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]\right).$$

↪**Proposition 3.11**: $\iint_{S_+} K \, \mathrm{d}\sigma = 2 \int_{\mathcal{C}} \kappa_{\mathcal{C}} \, \mathrm{d}s$

Proof. This follows from direct computation:

$$\begin{aligned}
\iint_{S_+} K \, \mathrm{d}\sigma &= \int_0^\ell \int_{\pi/2}^{(3\pi)/2} (K \circ \varphi)(s, \theta) \|\varphi_s \times \varphi_\theta\| \, \mathrm{d}\theta \, \mathrm{d}s \\
&= \int_0^\ell \int_{\pi/2}^{3\pi/2} -\cos\theta \kappa_\gamma(s) \, \mathrm{d}s \\
&= 2 \int_{\mathcal{C}} \kappa_{\mathcal{C}} \, \mathrm{d}s
\end{aligned}$$

■

↪**Proposition 3.12**: The map $N : S_+ \to \mathbb{S}^2$ is surjective and thus $\iint_{S^+} K \, \mathrm{d}\sigma \geq 4\pi$.

Proof. We give a heuristic argument. Fix $v \in \mathbb{S}^2$ and let $\mathcal{P}$ be the plane in $\mathbb{R}^3$ with normal v. Since S compact, we can translate $\mathcal{P}$ in space such that it doesn't intersect S anywhere. We can then translate $\mathcal{P}$ towards S until it first intersects it, at some point p;

this means $\mathcal{P}$ is tangent to S at p and thus $N(p) = v$. Moreover, by a second-derivative test type argument, $K(p) \geq 0$. Thus we have surjectivity.

For the "thus" statement, we can apply a change of variables to the integral in question and get

$$\iint_{S_+} K \, \mathrm{d}\sigma = \iint_{S_+} \det(\mathrm{d}N) \, \mathrm{d}\sigma \geq \iint_{N(S_+)} \mathrm{d}\sigma_{\mathbb{S}^2} = 4\pi,$$

where the inequality comes from the surjectivity (if N were a homeomorphism, we would have equality here). ■

↪**Lemma 3.2**: Suppose there exists a $v \in \mathbb{S}^2$ such that the *signed height function*

$$H_v : C \to \mathbb{R}, \quad w \mapsto w \cdot v$$

has exactly 2 local extrema. Then, C is unknotted.

Proof. Aw man. ■

↪**Proposition 3.13**: If C is knotted, then $N : S_+ \to \mathbb{S}^2$ is at least 2 to 1.

Proof. We prove the contrapositive. Suppose $N : S_+ \to \mathbb{S}^2$ is not 2 to 1. We know it is 1 to 1, so there must exist some $v \in \mathbb{S}^2$ and $p \in S_+$ such that $N(p) = v$, but $N(q) \neq v$ for all $q \neq v \in S_+$. Consider the height function H_v. In local coordinates,

$$(H_v \circ \gamma)(t) = \gamma(t) \cdot N(p)$$

so that

$$\frac{\mathrm{d}}{\mathrm{d}t}(H_v \circ \gamma)(t) = \dot{\gamma}(t) \cdot N(p).$$

This can only vanish if $\dot{\gamma}(t)$ orthogonal to $N(p)$. Ummm

■

Combining these results, the last proposition really tells us that

$$\iint_{S_+} K \, \mathrm{d}\sigma \geq 8\pi$$

by a similar argument to the last last proposition. Combining with the first proposition yields the result.

§3.5 Curvature under Local Isometries, and Gauss's and Bonnet's Theorems

Fix S a regular C^2 surface with orientation $N : S \to \mathbb{S}^2$ (which is C^1), and let $\varphi : U \to S$ a regular C^2 parametrization compatible with N (i.e. $N = \frac{\partial_u \varphi \times \partial_v \varphi}{\|\partial_u \varphi \times \partial_v \varphi\|}$). This means

$$\{\partial_u \varphi, \partial_v \varphi, N \circ \varphi\}$$

forms a positively-oriented ordered basis of $\mathbb{R}^3$. Akin to the Frenet frame equations, we will consider the derivatives of this frame. Recall that in local coordinates and in the $\{\partial_u \varphi, \partial_v \varphi\}$ basis,

$$-J_N = \begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \begin{pmatrix} e & f \\ f & g \end{pmatrix} =: \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

We compute first

$$\partial_u(N \circ \varphi) = -a_{11}\partial_u\varphi - a_{21}\partial_v\varphi$$
$$\partial_v(N \circ \varphi) = -a_{12}\partial_u\varphi - a_{22}\partial_v\varphi.$$

These constants we can find explicitly in terms of $E, F, G; e, f, g$. We similarly find

$$\partial^2_{uu}\varphi = \Gamma^1_{11}\partial_u\varphi + \Gamma^2_{11}\partial_v\varphi + eN$$
$$\partial^2_{uv}\varphi = \Gamma^1_{12}\partial_u\varphi + \Gamma^2_{12}\partial_v\varphi + fN$$
$$\partial^2_{vu}\varphi = \Gamma^1_{21}\partial_u\varphi + \Gamma^2_{21}\partial_v\varphi + fN$$
$$\partial^2_{vv}\varphi = \Gamma^1_{22}\partial_u\varphi + \Gamma^2_{22}\partial_v\varphi + gN,$$

where Γ^1_{11}, etc, are the appropriate scalar constants in this basis for the given vectors. Note that since $\varphi \in C^2$, $\Gamma^i_{jk} = \Gamma^i_{kj}$ (i.e., the middle two equations are equivalent). We call the coefficients, which we view as functions

$$\Gamma^i_{jk} : U \to \mathbb{R}, \quad 1 \leq i, j, k \leq 2$$

the *Christoffel symbols*. The lower indices indicate the derivatives of φ to which Γ^i_{jk} corresponds to (i.e. $(jk) = (11)$ means take two derivatives in the first coordinate i.e. u, etc), and the upper index indicates to which coefficient of the basis (1 for $\partial_u\varphi$, the first basis element, 2 for $\partial_v\varphi$, the second basis element) Γ^i_{jk} is multiplied to.

We can write the Christoffel symbols explicitly in terms of E, F, G and their derivatives. One computes

$$\partial_u E = 2\partial^2_{uu}\varphi \cdot \partial_u\varphi,$$
$$\partial_v E = 2\partial^2_{uv}\varphi \cdot \partial_u\varphi,$$

and similarly

$$\partial_u F = \partial^2_{uu}\varphi \cdot \partial_v\varphi + \partial_u\varphi \cdot \partial^2_{uv}\varphi$$
$$\partial_v F = \partial^2_{uv}\varphi \cdot \partial_v\varphi + \partial_u\varphi \cdot \partial^2_{vv}\varphi,$$

similarly for G. Combining with our equations above, we find

$$\partial^2_{uu}\varphi \cdot \partial_u\varphi = \Gamma^1_{11}E + \Gamma^2_{11}F = \frac{1}{2}\partial_u E$$
$$\partial^2_{uu}\varphi \cdot \partial_v\varphi = \Gamma^1_{11}F + \Gamma^2_{11}G = \partial_u F - \frac{1}{2}\partial_v E,$$

which can be written in matrix form as the expression

$$\begin{pmatrix} \Gamma^1_{11} \\ \Gamma^2_{11} \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \begin{pmatrix} \frac{1}{2}\partial_u E \\ \partial_u F - \frac{1}{2}\partial_v E \end{pmatrix},$$

where we recall that since S regular, the 1st fundamental form is invertible. One similarly computes (follow the exact same reasoning as above) for the other pairs of Christoffel symbols

$$\begin{pmatrix}\Gamma_{12}^1\\ \Gamma_{12}^2\end{pmatrix} = \begin{pmatrix}E & F\\ F & G\end{pmatrix}^{-1}\begin{pmatrix}\frac{1}{2}\partial_v E\\ \frac{1}{2}\partial_u G\end{pmatrix}, \quad \begin{pmatrix}\Gamma_{22}^1\\ \Gamma_{22}^2\end{pmatrix} = \begin{pmatrix}E & F\\ F & G\end{pmatrix}^{-1}\begin{pmatrix}\partial_v F - \frac{1}{2}\partial_u G\\ \frac{1}{2}\partial_v G\end{pmatrix}.$$

In short, we see that the Christoffel symbols are completely determined by the 1st fundamental form, and thus are preserved under local isometries.

Assume now S is C^3 and N is C^2, so that in particular $\partial_{uv}^2(\partial_u\varphi) = \partial_{vu}^2(\partial_u\varphi)$, $\partial_{uv}N = \partial_{vu}N$, etc.. Let us expand this first identity. The right-hand side can be written, using the Christoffel symbols from before,

$$\partial_v(\partial_{uu}^2\varphi) = \partial_v\Gamma_{11}^1\partial_u\varphi + \Gamma_{11}^1\partial_{vu}^2\varphi + \partial_v\Gamma_{11}^2\partial_v\varphi + \Gamma_{11}^2\partial_{vv}^2\varphi + \partial_v eN + e\partial_v N,$$

and similarly the left-hand side can be written

$$\partial_u(\partial_{vu}^2\varphi) = \partial_u\Gamma_{12}^1\partial_u\varphi + \Gamma_{12}^1\partial_{uu}^2\varphi + \partial_u\Gamma_{12}^2\partial_v\varphi + \Gamma_{12}^2\partial_{uv}^2\varphi + \partial_u fN + f\partial_u N.$$

We can (well we could, but we'll just do it in our heads…) rewrite $\partial_u N, \partial_v N, \partial_{uu}^2\varphi, \partial_{uv}^2\varphi, \partial_{vu}^2\varphi, \partial_{vv}^2\varphi$ in terms of our basis, and since these two terms must be equal, match the coefficients of each basis element.

For $\partial_u\varphi$:

$$\begin{aligned}
\partial_u\Gamma_{12}^1 + \Gamma_{12}^1\Gamma_{11}^1 + \Gamma_{12}^2\Gamma_{12}^1 - fa_{11} &= \partial_v\Gamma_{11}^1 + \Gamma_{11}^1\Gamma_{12}^1 + \Gamma_{11}^2\Gamma_{22}^1 - ea_{12}\\
\Rightarrow fa_{11} - ea_{12} &= \partial_u\Gamma_{12}^1 - \partial_v\Gamma_{11}^1 + \Gamma_{12}^2\Gamma_{12}^1 - \Gamma_{11}^2\Gamma_{22}^1\\
\Rightarrow \frac{feG - f^2F}{EG - F^2} - \frac{efG - egF}{EG - F^2} &= (\cdots)\\
\Rightarrow \frac{eg - f^2}{EG - F^2}F &= (\cdots)\\
\Rightarrow K\cdot F &= (\cdots)\\
\Rightarrow K &= \frac{\partial_u\Gamma_{12}^1 - \partial_v\Gamma_{11}^1 + \Gamma_{12}^2\Gamma_{12}^1 - \Gamma_{11}^2\Gamma_{22}^1}{F}.
\end{aligned}$$

With the dust settled, the punchline is that we have written K purely in terms of the Christoffel symbols and the first fundamental form. Since these are invariant under local isometries, this implies the Gaussian curvature is invariant under local isometries, assuming $F \neq 0$ (since we divided by F at the end). However, if F is 0, we can similarly match the coefficients for $\partial_v\varphi$ and compute that

$$\boxed{K = \frac{\Gamma_{11}^1\Gamma_{12}^2 - \Gamma_{12}^1\Gamma_{11}^1 + \Gamma_{11}^2\Gamma_{22}^2 - \Gamma_{12}^2\Gamma_{12}^2 + \partial_v\Gamma_{11}^2 - \partial_u\Gamma_{12}^2}{E},}$$

and since E is always nonzero (for regular surfaces), this is always a valid expression for K in terms of the first fundamental form. This is known as the *Gauss equation*, which proves

↪**Theorem 3.5** (Gauss's Theorema Egregium): The Gaussian curvature of a C^3 surface is invariant under local isometries.

Exercise 3.8: Suppose $E \equiv 1$ and $F \equiv 0$. Show that $K = -\frac{\partial^2_{uu}\left(\sqrt{G}\right)}{\sqrt{G}}$.

↪**Corollary 3.5**: If S is locally isometric to a plane (i.e. there exists a local parametrization such that $\mathrm{d}s^2 = \mathrm{d}u^2 + \mathrm{d}v^2$), then $K \equiv 0$.

Similarly, if S is locally isometric to $\mathbb{S}^2$, then $K \equiv 1$.

Finally, we can compare the coefficients of N, from which we find the relationship

$$\Gamma^1_{11} f + \Gamma^2_{11} g + \partial_v e = \Gamma^1_{12} e + \Gamma^2_{12} f + \partial_u f \rightsquigarrow \quad \text{1st "Mainardi-Codazzi Equation".}$$

Similarly, we can compare the coefficients of the basis elements upon expanding the relationships $\partial^2_{uv}\partial_v\varphi = \partial^2_{vu}\partial_v\varphi$ and $\partial^2_{uv}N = \partial^2_{vu}N$ to find a family of 9 compatibility equations involving the first and second fundamental forms.

↪**Theorem 3.6** (Bonnet): Suppose $E, F, G, e, f, g : V \to \mathbb{R}$ are smooth functions on $V \subset \mathbb{R}^2$ open, with $E, G, EG - F^2 > 0$. If they also satisfy the aforementioned compatibility equations, there exists $U \subset V$ open and a regular smooth parametrization $\varphi : U \to \mathbb{R}^3$ such that

$$\mathrm{I}_\varphi = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad \mathrm{II}_\varphi = \begin{pmatrix} e & f \\ f & g \end{pmatrix}$$

and φ is unique up to rigid motions.

Remark 3.14: We use "smooth" to avoid digging into the exact conditions of the compatibility equations.

Remark 3.15: We can use Gauss's formula to extend the definition of Gaussian curvature for regular C^3 surfaces in $\mathbb{R}^n$. This begins by defining it analogously to above for a local parametrization, then by showing that this definition is invariant under change of parametrization.

§4 Appendix

§4.1 Auxiliary Results

↪**Proposition 4.1**: For $u, v \in \mathbb{R}^3$, $\|u \times v\|$ is the area of the parallelogram with side u, v.

↪**Proposition 4.2**: Let $M \in \mathbb{R}^{3\times 3}$ with column vectors M_1, M_2, M_3. Then

$$\det(M) = (M_1 \times M_2) \cdot M_3 = (M_2 \times M_3) \cdot M_1 = (M_3 \times M_1) \cdot M_1.$$

§4.2 Summary of Functions

Formulae for Curves

Let $\gamma : I \to \mathbb{R}^n$ be a parametrized curve, and let $\tilde{\gamma} = \gamma \circ s$ be an arc-length reparametrization of γ.

Formula for:	γ	$\tilde{\gamma}$	$n = 2$ specific?	$n = 3$ specific?
curvature ‡	$\frac{\|\ddot{\gamma}^{\perp}\|}{\|\dot{\gamma}\|^2}$	$\|\ddot{\tilde{\gamma}}\|$		$\frac{\|\dot{\gamma}\times\ddot{\gamma}\|}{\|\dot{\gamma}\|^3}$
signed curvature ‡	$\frac{\ddot{\gamma}\cdot\dot{\gamma}^*}{\|\dot{\gamma}^3\|}$		$\frac{\dot{x}\ddot{y}-\ddot{x}\dot{y}}{(\dot{x}^2+\dot{y}^2)^{3/2}}$	N/A
tangent †	$\frac{\dot{\gamma}}{\|\dot{\gamma}\|}$	$\dot{\tilde{\gamma}}$		
normal ‡	$\frac{\ddot{\gamma}^{\perp}}{\|\ddot{\gamma}^{\perp}\|}$	$\frac{\ddot{\tilde{\gamma}}}{\|\ddot{\tilde{\gamma}}\|}$		
length †	$\int_{t_0}^{t} \|\dot{\gamma}\| \, \mathrm{d}s$	$t - t_0$		
osculating circle ‡	radius $= \frac{1}{\kappa}$ center $= \gamma + \frac{N}{\kappa}$			N/A
evolute ‡	$\gamma + \frac{N}{\kappa}$			N/A
theta ‡	$\theta_0 + \int_{t_0}^{t} \kappa_\gamma \, \mathrm{d}u$	$\tilde{\theta}_0 + \int_{s_0}^{s} \kappa_{\tilde{\gamma}}^{\pm} \, \mathrm{d}u$		N/A
binormal $\overset{*}{‡}$		$B = T \times N$	N/A	
torsion $\overset{*}{‡}$	$\frac{\dot{\gamma}\times\ddot{\gamma}}{\|\dot{\gamma}\times\ddot{\gamma}\|^2} \cdot \dddot{\gamma}$	$-\dot{B} \cdot N$	N/A	

† = requires regular, ‡ = requires C^2, $\overset{*}{‡}$= requires C^3, N/A= doesn't exist.